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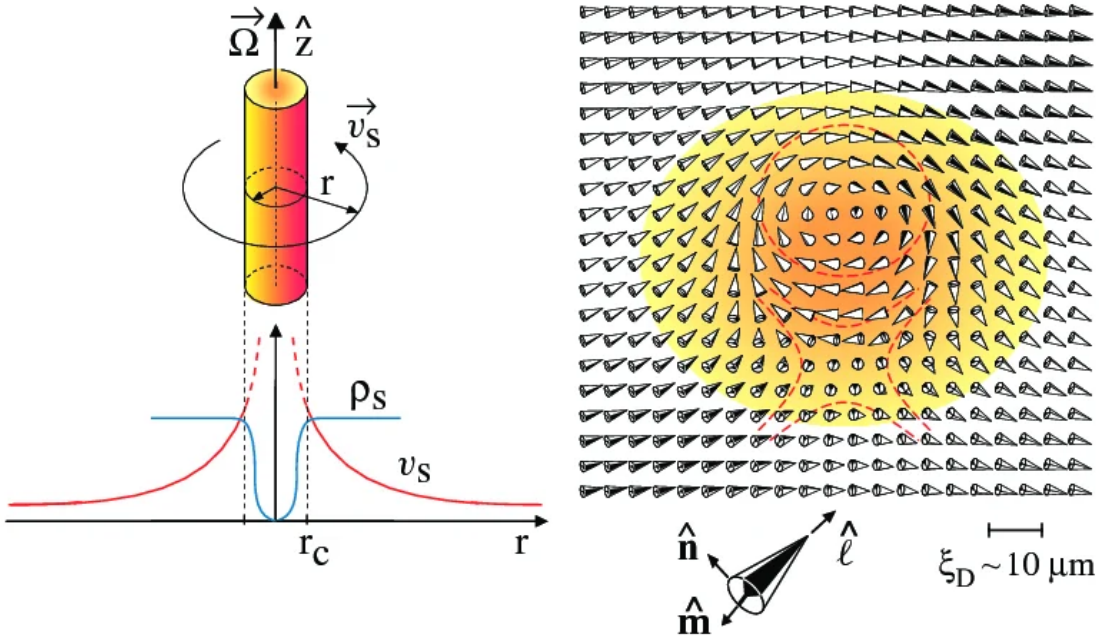

(Left) Macroscopic structure of quantized vortex line in He superfluids. The core radius rc is on the order of the superfluid coherence length in 4 He-II (ξ ~ 0.1 nm) and 3 He-B (ξ 10 nm), but in 3 He-A the length scale is the healing length of the dipolar spin-orbit interaction (ξ D 10 μm). (Right) Orbital order parameter texture of the soft core of the double-quantum vortex in 3 He-A in magnetic field. The cones indicate the local direction and rotation of the orbital order parameter triad of unit vectorsîvectorsî, ˆ m, ˆ n. The topological winding number of thê l texture is n = 2. The texture is nonaxisymmetric: it is composed of a circular half, or meron, and a hyperbolic meron, each with 2π circulation.

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Dynamics of vortices and interfaces in superfluid 3He

Article Full-text available

Jun 2006

A. P. Finne · V. B. Eltsov · R. Hanninen · [...] · Grigory Volovik

Rapid new developments have occurred in superfluid hydrodynamics since the discovery of a host of unusual phenomena which arise from the diverse structure and dynamics of quantized vortices in 3He superfluids. These have been studied in rotating flow with NMR measurements which at best provide an accurate mapping of the different types of topologic...

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Contexts in source publication

Context 1

... the magnetic field of the NMR measurements the continuous vortex is deformed and its structure is nonaxisymmetric, see Fig. 1. However, its topology is robust to deformations, and the circulation remains the same: dr · v s = 2κ. It is important to note that, since even in the soft-core region the order parameter retains its bulk structure, the core size of the continuous 3 He-A vortex is not set by the coherence length ξ 10 nm of the superfluid state. ...

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Context 2

... the two halves of the skyrmion texture, the circular and hyperbolic merons (Fig. 1) as

basic building blocks, other structures of continuous vorticity can be formed. An example are the various continuous periodic vortex textures in zero or low magnetic field [35,36]. Another important structure is the vortex sheet [37,38] which competes for living space with the double-quantum vortex line. A concise lexicon of these ...

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Context 3

... critical velocity approaches zero, although it apparently always remains finite, and all vortex quanta will enter from along one of the two connection lines of the soliton sheet with the cylinder wall. This means that the emerging vortices are not lines but a periodic chain of circular and hyperbolic vortex quanta stacked within a soliton sheet (Fig. 1). This structure is called a vortex sheet [37]. It has the lowest critical velocity of all A-phase vortex structures and also the fastest dynamic response [62]. Numerical calculations of the flow instability of various different one-dimensional initial textures [63] show that semi-quantitative agreement with the measured variation ...

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Context 4

... this regime the parameter Γ ∝ T 3 [95,96,97]. For simplicity, we choose a situation Figure 1 0. Sketch of the imaginary and real parts of the frequency ω(k) of the critical ripplon (when k = k 0) at the interface between 3 He-A and 3 He-B. ...

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Context 5

... when the expression under the square root becomes negative. The ripplon spectrum then acquires an imaginary part with both plus and minus signs. Thus at the threshold velocity of Eq. (31) the perturbation grows exponentially in time. A sketch of the imaginary and real parts of the frequency of the critical ripplon mode (with k = k 0) is shown in Fig. 10. In the case of the superfluid instability we have to consider the influence of the term with the friction parameter Γ in Eq. 35. When the imaginary part Im ω(k) crosses zero as a function of U B , the attenuation of the ripplon modes is transformed to amplification. This occurs first for ripplons with the wave vector given by Eq. ...

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Context 6

... is also possible here to identify a similarity with the Kelvin-wave instability of an

isolated vortex line in applied flow. In the T → 0 limit, when both α → 0 and α ′ → 0, the instability of a vortex line in externally applied superflow towards the formation of Kelvin waves resembles the AB interface instability. For the unstable modes in Fig. 10 with Im ω > 0, the real part of the spectrum is negative, Re ω < 0. Similarly, for a vortex of finite length L the wave-number is constrained from below, k > k 0 = 2π/L, and the instability forms at a critical velocity v c ~ ˜ κk 0 . This v c does not depend on the friction parameter α, whose role is to provide the dissipation from the ...

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Context 7

... superfluid KH instability of the AB interface is a new physical phenomenon with wide-ranging ramifications, as discussed in Secs. 3.1 -3.5. It also provides a whole new set of tools for further measurements on vortex dynamics. The standard KH measurement is that of the critical rotation velocity Ω c shown in Fig. 11. Here the NMR absorption plotted on the vertical scale monitors the number of rectilinear vortices in the B phase while the rotation velocity Ω on the horizontal axis is slowly increased. Temperature T and barrier current I b are kept constant during the measurement. The first discontinuous jump in NMR absorption marks Ω c . This is ...

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Context 8

... calibrated to provide the exact number of vortices which is transferred across the AB interface in each single instability event. Their number ∆N is denoted next to each instability event. We see that ∆N is a small random number, which after many similar measurements is found to be between 3 -30 with an average of ~ 10 [104]. The measurement in Fig. 11 is performed at high temperatures where each of the ∆N vortex loops rapidly develops into a rectilinear B-phase vortex line, after they have been tossed as a tight bundle (Top) A step-like increase in absorption is observed in the Larmor region of the NMR line shape when the rotation velocity Ω is slowly increased and the critical ...

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Context 9

... the AB interface. From such measurements at different temperatures, Ω c has been collected to provide the curves at different values of I b seen in Fig. 9. In the measurement of Fig. 11 the sample might be in either the BA or BAB configurations. In the setup of Fig. 7 the magnetic field distribution along the vertical axis is almost

symmetric with respect to the middle of the sample. Also the end plates of the sample cylinder and the NMR pick-up coils are at the same distance from the AB interfaces in both B-phase ...

View in full-text

Context 10

... of Ω c , the rotation velocity corresponding to the first instability event, are identical for the top and bottom B-phase sections in the BAB configuration. Nevertheless, the events at the two AB interfaces in the BAB configuration are not correlated and ΔN differs evidently randomly from one event to the next. Therefore, as can be seen from Fig. 11, subsequent instability events at the two AB interfaces get soon out of step, when Ω is increased above Ω c , and occur in the end randomly, but such that the critical counterflow velocity remains fixed: v c ≈ Ω c R. The reason for this is that vortices play no direct role in the instability condition of Eq. (30): Even though the ...

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Context 11

... end randomly, but such that the critical counterflow velocity remains fixed: v c ≈ Ω c R. The reason for this is that vortices play no direct role in the instability condition of Eq. (30): Even though the vortices in the A-phase vortex layer covering the AB interface are as closely packed next to each other as the soft core diameter allows (see Fig. 13 right), they do not determine the instability. Instead it is the stability of the interface itself in the tangential flow according to Eq. (30). A schematic illustration is shown in Fig. 12 of the process in which ΔN vortex loops might be transferred across the AB interface. When the instability threshold is reached while increasing Ω, ...

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Context 12

... Even though the vortices in the A-phase vortex layer covering the AB interface are as closely packed next to each other as the soft core diameter allows (see Fig. 13 right), they do not determine the instability. Instead it is the stability of the interface itself in the tangential flow according to Eq. (30). A schematic illustration is shown in Fig. 12 of the process in which ΔN vortex loops might be transferred across the AB interface. When the instability threshold is reached while increasing Ω, surface waves with wave vector k 0 = F/σ begin to form on the AB interface. At this stage the ΔN vortices correspond to the A-phase ...

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Context 13

... (μm) 579 λ/(2a s) P = 29 bar, T/T c = 0.56-0.58 circulation quanta which reside in one of the corrugations of the wave where it is about to become the deepest and most dominant trough. In the lower row of diagrams in Fig. 12 the possible sequence of events is depicted which might then take place. The trough becomes a potential minimum for vortices, which pushes the vortices even deeper in this well. Ultimately, as the trough gets distorted, a Magnus force starts to act on the vortices. The interface motion is highly damped and it moves faster than the ...

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Context 14

... further illustration of the KH measurement and its agreement with Eq. (30) is shown in Fig. 13. This plot is generated by measuring Ω c in similar manner as in Fig. 11, by increasing rotation to Ω c at different values of I b , keeping temperature constant. Alternatively, the measurement can be performed by decreasing I b at constant Ω, so that the critical curve in Fig. 13 is traversed horizontally from right to left. Since the ...

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Context 15

... further illustration of the KH measurement and its agreement with Eq. (30) is shown in Fig. 13. This plot is generated by measuring Ω c in similar manner as in Fig. 11, by increasing rotation to Ω c at different values of I b , keeping temperature constant. Alternatively, the measurement can be performed by decreasing I b at constant Ω, so that the critical curve in Fig. 13 is traversed horizontally from right to left. Since the measurement is at constant temperature, it is also performed at constant ...

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Context 16

... of the KH measurement and its agreement with Eq. (30) is shown in Fig. 13. This plot is generated by measuring Ω c in similar manner as in Fig. 11, by increasing rotation to Ω c at different values of I b , keeping temperature constant. Alternatively, the measurement can be performed by decreasing I b at constant Ω, so that the critical curve in Fig. 13 is traversed horizontally from right to left. Since the measurement is at constant temperature, it is also performed at constant value of H AB (T, P). The result demonstrates the dependence of the instability condition in Eq. (30) on the restoring force

in Eq. (26), owing to the dependence of ∇H on l b . When l b is changed, the AB ...

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Context 17

... the barrier magnet to a location where the field approximates the equilibrium transition field, H(r, z) ≈ H AB . Because of the large curvature in the solenoidal field and the surface tension σ of the AB interface, the equality is only approximately followed. At large l b the interface is flat, as seen in Fig. 7, and Ω c changes only slowly in Fig. 13 Fig. 13 correspond already to a configuration where the A-phase volume is not a complete layer between two independent B-phase sections, but where the AB interface separates out a torus-shaped A-phase volume around the cylinder wall, with a B-phase channel connecting the top and bottom sections of the sample [105]. Surprisingly, even ...

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Context 18

... magnet to a location where the field approximates the equilibrium transition field, H(r, z) ≈ H AB . Because of the large curvature in the solenoidal field and the surface tension σ of the AB interface, the equality is only approximately followed. At large l b the interface is flat, as seen in Fig. 7, and Ω c changes only slowly in Fig. 13 Fig. 13 correspond already to a configuration where the A-phase volume is not a complete layer between two independent B-phase sections, but where the AB interface separates out a torus-shaped A-phase volume around the cylinder wall, with a B-phase channel connecting the top and bottom sections of the sample [105]. Surprisingly, even in this ...

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Context 19

... high temperatures above 0.6 T c the dynamical behavior of vortices in 3 He-B is regular, i.e. their number does not increase during a time-dependent process. This is seen from the fact that single-vortex steps are observed in Figs. 3 and in the KH measurement of Fig. 11 the number of vortices ∆N formed in each instability event fits within the distribution expected on the basis of Fig. 13 (right). Some time ago it was recognized that at lower temperatures single-vortex processes are not observed in large applied flow [46]; instead, a large number of vortices is suddenly formed so that the final state ...

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Context 20

... dynamical behavior of vortices in 3 He-B is regular, i.e. their number does not increase during a time-dependent process. This is seen from the fact that single-vortex steps are observed in Figs. 3 and in the KH measurement of Fig. 11 the number of vortices ΔN formed in each instability event fits within the distribution expected on the basis of Fig. 13 (right). Some time ago it was recognized that at lower temperatures single-vortex processes are not observed in large applied flow [46]; instead, a large number of vortices is suddenly formed so that the final state in rotation appears to include close to the equilibrium number of vortex lines. More recently, a consistent explanation of this ...

View in full-text

Context 21

... as a function of temperature, it is now understood that mutual friction divides the dynamics in 3 He-B into a high-temperature regime with regular vortex-number conserving motion, and a low-temperature regime where superfluid turbulence becomes possible. This hydrodynamic transition is unusually abrupt as a function of temperature, as seen in Fig. 14. This is the reason why it was first mistakenly interpreted to signal a strongly temperature-dependent new critical velocity in Ref. [46]. The applied flow velocity, the counterflow velocity v n − v s , is an important factor in this transition: at low velocity a single vortex has been observed to be in stable precessing motion for an ...

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Context 22

... at high temperatures only the KH vortices are generated while at low temperatures the equilibrium vortex state is obtained after the instability event. The KH critical velocity as a function of temperature is a continuous smooth curve across this division line and continues to follow the calculated dependence. The same features are illustrated in Fig. 13 which is measured at a temperature close to the division line. A surprising and characteristic property of these plots is that there are essentially no data points with an arbitrary intermediate number of vortices, not even in the transition regime. The conclusion from these measurements is that at temperatures below the division line ...

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Context 23

... data points with an arbitrary intermediate number of vortices, not even in the transition regime. The conclusion from these measurements is that at temperatures below the division line a short turbulent burst follows the KH event, after the closely packed bundle

of vortex loops has arrived across the AB interface into the vortex-free B-phase flow (Fig. 12). The turbulent burst generates the vortices needed to reach almost the equilibrium number of vortices for the B-phase flow. The propagation of these vortices along the long rotating sample as a vortex front followed by a twisted vortex bundle is discussed in Sec. 5. A compelling argument for the interpretation in terms of a turbulent ...

View in full-text

Context 24

... since the KH instability was originally the most convincing case of vortex injection and of the turbulent burst, with very particular properties, we describe this example here in more detail. Figure 1 5 shows a close-up of the transition region as a function of temperature. The number of vortex lines N is plotted, normalized by the equilibrium number N eq , after a single event of KH injection. ...

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Context 25

... here in more detail. Figure 1 5 shows a close-up of the transition region as a function of temperature. The number of vortex lines N is plotted, normalized by the equilibrium number N eq , after a single event of KH injection. At high temperatures the injection leads always to a small number of rectilinear lines, as expected for the KH instability (Fig. 11). Below 0.6 T c the number of lines, after the system has settled down, is very different: the injection results in the almost complete removal of counterflow. The two regimes are separated by an abrupt transition at T on ≈ 0.58 T c which has a narrow width of ΔT on ≈ 0.04 T c . We attribute the final state at temperatures below T on as ...

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Context 26

... transition from Fig. 15 is shown as a phase diagram in Fig. 16, plotted in terms of the B-phase counterflow velocity v = ΩR and temperature T . In this diagram each data point represents a KH injection measurement, accumulated with different settings of the externally controlled parameters, so that as wide a variation as possible is obtained for the critical ...

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Context 27

... transition from Fig. 15 is shown as a phase diagram in Fig. 16, plotted in terms of the

B-phase counterflow velocity v = ΩR and temperature T . In this diagram each data point represents a KH injection measurement, accumulated with different settings of the externally controlled parameters, so that as wide a variation as possible is obtained for the critical rotation velocity Ω c and temperature T ...

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Context 28

... in terms of the B-phase counterflow velocity v = ΩR and temperature T . In this diagram each data point represents a KH injection measurement, accumulated with different settings of the externally controlled parameters, so that as wide a variation as possible is obtained for the critical rotation velocity Ω c and temperature T . Each marker in Fig. 16 thus indicates a KH injection event into vortex-free counterflow with some parameter values which are not of interest in this context. What we are interested in here is the division in filled and open symbols: events followed by a turbulent burst are again marked with filled symbols () while events which only lead to a few rectilinear ...

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Context 29

... similar to Fig. 16 were also measured at 34 and 10 bar pressures [108]. In Fig. 17 the results are summarized. We assume that the transition is velocity independent in the measured range of velocities, and compile the data to show the probability of the transition between regular and turbulent cases as a function of temperature. The fitted Gaussian ...

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Context 30

... similar to Fig. 16 were also measured at 34 and 10 bar pressures [108]. In Fig. 17 the results are summarized. We assume that the transition is velocity independent in the measured range of velocities, and compile the data to show the probability of the transition between regular and turbulent cases as a function of temperature. The fitted Gaussian distributions give a narrow half-width of σ ≈ 0.03 T c at all ...

View in full-text

Context 31

... condition is independent of extrinsic quantities such as the counterflow velocity or the characteristic dimension R of the sample. The temperature dependence of ℜ for 3 He-B

and He-II is explicitly shown in Fig. 5. The above dimensional arguments on the transition between regular and turbulent vortex dynamics are in agreement with the results in Fig. 16, where the boundary between the laminar and turbulent regimes was also found to be at q ~ 1. Thus the velocity independent parameter $\mathfrak{R}$ = 1/q, which controls the transition to turbulent flow in superfluids, plays the same role as the velocity dependent Reynolds number Re = U R/v in classical ...

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Context 32

... KH injection a tight bundle of many small vortex loops (Fig. 21) is transferred across the AB interface into vortex-free B-phase flow. How is a turbulent burst started from such a vortex bundle? In this section we discuss the initial phase of vortex multiplication in applied flow using a simple phenomenological model from Ref. [112] which is constructed in the spirit of the Vinen equation for ...

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Context 33

... ε k is the energy flux in k-space, which is constant in the inertial range of k, where viscous dissipation can be neglected, ε k = ε. Here, we study what is the outcome of the turbulent instability in a superfluid in different regimes of Reynolds parameters $\mathfrak{R}$ = 1/q and Re s . We show that in the intermediate range of Reynolds numbers 1/q (Fig. 18) the turbulent instability leads to a state of developed turbulence which is closer to its classical analogue: It exhibits a Richardson-like cascade [119] and is different from the Vinen state of ...

View in full-text

Context 34

... cascade is then transformed to the Kelvin-wave cascade [2] of isolated vortices. In the second scenario the turbulent state is completely reconstructed and the Vinen state discussed in Sec. 4.4 and Refs. [113,114] emerges. This state contains a single scale ℓ = κ/U and thus no cascade. A possible phase diagram of the turbulent states is shown in Fig. 18. The connection of this phase diagram with the flow states observed in various experiments on superfluid 4 He-II and 3 He-B is discussed in Ref. ...

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Context 35

... do seed vortices injected in applied counterflow undergo a transition to turbulence? A superficial answer can be given by inspecting the diagram in Fig. 19. Here vortex-free flow is generated by subjecting the superfluid sample to uniform rotation. This is a metastable state of high energy where the superfluid fraction does not participate in the rotation. As long as the resulting maximum counterflow velocity remains below the limit of spontaneous vortex formation of the particular sample ...

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Context 36

... mechanisms. The injection is carried out at constant external con- ditions while the rotation velocity Ω is suddenly incrementally increased by ΔΩ ≈ 0.05 rad/s across the critical KH value Ω c (T, P). At temperatures above T on such injection generates a limited number of vortices in the B phase and consequently the staircase pattern in Fig. 11 can be displayed, by triggering multiple instability events one after the other. Below T on the staircase pattern is not obtained, since already the very first injection sends the sample in the equilibrium vortex state and removes essentially all applied counterflow. The measured signature from the injection in the turbulent ...

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Context 37

... events one after the other. Below T on the staircase pattern is not obtained, since already the very first injection sends the sample in the equilibrium vortex state and removes essentially all applied counterflow. The measured signature from the injection in the turbulent temperature regime is shown in Fig. 20. The radical difference from Fig. 11 is evident: The dynamic formation of new vortices in a turbulent burst increases the vortex number immediately to N eq ...

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Context 38

... evolution following a triggered injection event can be monitored by recording the NMR absorption height as a function of time either at the location of the counterflow peak, as is done in Fig. 20 to measure the reduction in the macroscopic counterflow, or in the region of the Larmor edge, as is done in Fig. 11 to exhibit a signal which is generated more directly by vortices. In both regions the NMR line shape arises from the interplay of vortex-free counterflow and of vortices via their orientational influence on the order parameter texture, as explained in the ...

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Context 39

... injection produces the transition to turbulence at the highest onset tem- perature. The reason is interpreted to be the initial configuration in which the seed vortices end up on the B-phase side of the interface: As sketched in Fig. 21, they form a tightly packed bundle of many roughly parallel vortex loops, as estimated in Fig. 13. Apparently, in the applied B-phase flow they immediately start interacting turbulently, when the amplitudes of Kelvin wave excitations are not over-damped. This leads instantaneously to a turbulent burst and gives the highest onset ...

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Context 40

... injection produces the transition to turbulence at the highest onset tem- perature. The reason is interpreted to be the initial configuration in which the seed vortices end up on the B-phase side of the interface: As sketched in Fig. 21, they form a tightly packed bundle of many roughly parallel vortex loops, as estimated in Fig. 13. Apparently, in the applied B-phase flow they immediately start interacting turbulently, when the amplitudes of Kelvin wave excitations are not over-damped. This leads instantaneously to a turbulent burst and gives the highest onset temperature T on Seed loop injection at the AB interface is followed by a turbulent burst, rapid ...

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Context 41

... is signalled as an incremental increase in the counterflow peak heights of both signals in the main panel. double-quantum vortex vortex layer coating AB interface Figure 21. Sketch of the initial vortex loop configuration in KH injection. ...

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Context 42

... vortex free, except for the seed loops which have escaped across the AB interface in a KH instability event. Initially they are contained in a tight bundle which is oriented radially in the cylinder, starting from R eff at the AB interface and ending at the cylindrical wall. The bundle has roughly a diameter ~ 0.1 -0.2 mm (corresponding to λ/2 in Fig. 13 center), a length ~ 0.4 mm (corresponding to R − R eff in Fig. 11), and contains on an average 10 vortex loops with a spacing as ~ 15 -25 μm (Fig. 13 right). In the turbulent temperature regime the vortices within the bundle are rapidly destabilized by Kelvin wave

excitations, owing to the applied flow v = vn − vs = Ω × r and the interactions ...

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Context 43

... AB interface in a KH instability event. Initially they are contained in a tight bundle which is oriented radially in the cylinder, starting from R eff at the AB interface and ending at the cylindrical wall. The bundle has roughly a diameter ~ 0.1 -0.2 mm (corresponding to λ/2 in Fig. 13 center), a length ~ 0.4 mm (corresponding to R − R eff in Fig. 11), and contains on an average 10 vortex loops with a spacing as ~ 15 -25 μm (Fig. 13 right). In the turbulent temperature regime the vortices within the bundle are rapidly destabilized by Kelvin wave excitations, owing to the applied flow v = vn − vs = Ω × r and the interactions between the loops. which, at least at higher flow ...

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Context 44

... bundle which is oriented radially in the cylinder, starting from R eff at the AB interface and ending at the cylindrical wall. The bundle has roughly a diameter ~ 0.1 -0.2 mm (corresponding to λ/2 in Fig. 13 center), a length ~ 0.4 mm (corresponding to R − R eff in Fig. 11), and contains on an average 10 vortex loops with a spacing as ~ 15 -25 μm (Fig. 13 right). In the turbulent temperature regime the vortices within the bundle are rapidly destabilized by Kelvin wave excitations, owing to the applied flow v = vn − vs = Ω × r and the interactions between the loops. which, at least at higher flow velocities, turns out to be independent of the flow velocity. In contrast such transitions to ...

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Context 45

... stochastic process. This means that at a flow velocity ΩR ~ v cni the number of rings obtained from a given neutron absorption event can be anything from zero up to the maximum limit i. At high temperatures T > T on , each extracted ring evolves independently to a rectilinear vortex line. These can be individually counted in similar fashion as in Fig. ...

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Context 46

... the injected vortices is more favourable for interactions between the expanding seeds. Eventually at the lowest temperatures ~ 0.30 T c a vortex injection of any kind will always

lead to turbulence. Overall the most unexpected new result is the fact that in the limiting case of a direct transition to turbulence, like Kelvin-Helmholtz injection in Fig. 16, the threshold becomes velocity independent, as can also be concluded from numerical simulations and theoretical models. These models now show that in the theory of homogeneous developed superfluid turbulence the Kolmogorov-Richardson cascade is modified by the influence of mutual ...

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Context 47

... provides a mapping of the spatial variation of the global order parameter texture. This texture is affected by the presence or absence of vortices. Thus indirectly from the changes in the global texture one can infer information about vortices, up to the point that in certain situations the vortices can be detected individually, as seen in Figs. 3 and 11. This is different from the A phase where NMR absorption is sensitive to the local order parameter distribution, in particular in locations where its alignments differ from the spin-orbit equilibrium configuration. Here each type of topological defect leaves a characteristic satellite peak in the absorption spectrum [143]. The ...

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Context 48

... general, metastable states containing a cluster of any number of vortices N between zero and the equilibrium number N eq can be observed in rotating 3 He-B (Fig. A1). In such a case, the magnitude of the azimuthal counterflow is v(r) = 0, 0 < r < R c , Ωr − ΩR 2 c /r, R c < r < R, (A.9) where the cluster radius R c = R N/N eq . The number of vortex lines in a given cluster can be determined by analyzing the counterflow peak height, either by comparing to numerically calculated equilibrium textures ...

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... These systems allow for the construction of analogies that mimic quantum field theory in curved spacetimes, enabling the simulation of phenomena such as Hawking radiation [6], 'trans-Planckian modes' [7] and their prospective detection in analogue platforms [8][9][10], as well as Bekenstein-Hawking entropy [11], among others [2,12]. Notable realizations include experiments involving vortices [13][14][15][16][17][18][19], superfluid helium [18, 20] , Bose-Einstein condensates [21,22], and related media. ...

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We present a geometrical formulation of nonlinear electrodynamics by expressing its principal symbol as an optical metric-induced object. Under the assumption of no birefringence, we show that the evolution of linear perturbations can be reca…

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... mechanisms: the Landau instability of ripplons at the interface of the A and B phases of superfluid 3 He (refs. 15, 17) and the point vortex array instability in scalar superfluids of bosonic and fermionic gases 23,24 . Therefore, the exploration of the quantum counterpart of KHI has been

limited. ...

... The interface displays a flutter-finger pattern that rolls up and enters a turbulence regime at later times. It can be found in various fluids with different length scales, such as oceans 13 , the cloud bands of Jupiter 14 and even in a quantum fluid [15][16] [17] [18][19][20][21] . In the KHI of immiscible multi-component superfluids, a universal (classical) regime exists 21 and displays the flutter-finger pattern through the KHI. ...

Stable singular fractional skyrmion spin texture from the quantum Kelvin–Helmholtz instability

Article | Full-text available

Aug 2025 · NAT PHYS

SeungJung Huh · Wooyoung Yun · Gabin Yun · Samgyu Hwang · Jae-yoon Choi

Topological defects give rise to non-trivial excitations—such as solitons, vortices and skyrmions—characterized by integer-valued topological charges. However, their classification becomes difficult when singularities are present in the order…

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... The system involves two complex scalar opera-torsÔ I which carry the same charge e under a global U (1) symmetry. 6 Below a critical temperature T c , the scalar operators develop a nonzero expectation value spontaneously breaking the U (1) symmetry and driving the system into a superfluid phase. The holographic system in (3+1)-dimensional asymptotically anti-de Sitter 6 Thus there will be only one conserved density, which is a special limit of the hydrodynamic and GPE analyses of the previous sections. ...

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one conserved density, which is a special limit of the hydrodynamic and GPE analyses of the previous sections. ...

Uncovering thermodynamic origin of counterflow and coflow instabilities in miscible binary superfluids

Preprint | Full-text available

Nov 2024

Yuping An · Blaise Gouteraux · Li Li

In this paper, we explore instabilities in binary superfluids with a nonvanishing relative superflow, particularly focusing on counterflow and coflow instabilities. We extend recent results on the thermodynamic origin of finite superflow…

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... For example, considerable attention has been devoted to the formation of von Kármán vortex streets in superfluid flows past obstacles [22][23][24], to the presence of boundary layers around the surfaces of moving objects [25], and to the Rayleigh-Taylor instability at the interface between two immiscible BECs [26][27][28][29]. As regards the Kelvin-Helmholtz instability, its first study in a superfluid system dates back to twenty years ago [30], where it was observed at the interface between 3 He − A and 3 He − B (see also the more recent Ref. [31]). Since then, a few theoretical and experimental works have followed, investigating its occurrence at the interface between a superfluid and a normal fluid [32,33], or between two different superfluids [34][35][36][37][38][39][40][41]. ...

... Comparing Eqs. (31) and (21), it is clear that the kinetic term T ′ a = πħ hn a N v j=1 (R 2 2 − r 2 j)θ ′ j is formally unaltered, while the potential term is modified as follows: ...

Suppression of the superfluid Kelvin-Helmholtz instability due to massive vortex cores, friction and confinement

Article | Full-text available

Sep 2024

Matteo Caldara · Andrea Richaud · Massimo Capone · Pietro Massignan

We characterize the dynamical instability responsible for the breakdown of regular rows and necklaces of quantized vortices that appear at the interface between two superfluids in relative motion. Making use of a generalized point-…

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... 26,27, the instability appears as the departure from the initial ordered vortex configuration with characteristic rates given by such as superfluid and normal phases of the same liquid 15,16 , and in binary Bose-Einstein condensates 17-20 (BECs). Experimental observations are limited to the interface between the A and B phases of 3 He in a rotating cryostat 21, 22 . More recently, the appearance of streets of quantized vortices in rapidly rotating atomic BECs has been associated with Kelvin-Helmholtz dynamics 23 . ...

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... Our observations establish atomic Fermi superfluids as a versatile laboratory for quantum fluid dynamics experiments. We overcome typical difficulties that affect experiments with helium quantum fluids 21, 22 , especially in terms of single-vortex detection and reconstruction of vortex trajectories 28 . ...

Connecting shear flow and vortex array instabilities in annular atomic superfluids

Article | Full-text available

Mar 2024 · NAT PHYS

D. Hernández-Rajkov · N. Grani · F. Scazza · G. Del Pace · G. Roati

At the interface between two fluid layers in relative motion, infinitesimal fluctuations can be exponentially amplified, inducing vorticity and the breakdown of laminar flow. While shear flow instabilities in classical fluids have been…

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... Such interfaces were first discussed in the context of domain walls in the early universe [1][2][3], where they may form the termination points of cosmic strings [4], and later in brane models in superstring theory [5][6][7]. They appear universally across many areas of physics, from the A−B phase boundary in superfluid liquid 3 He [8][9][10] [11] [12][13], via atomic Bose-Einstein condensates (BECs) [14][15][16][17][18][19], to quantum chromodynamics [20][21][22]. The different order-parameter symmetries imply that the bulk medium on either side of the interface supports different families of topological defects and textures, which therefore cannot cross the interface unchanged. ...

... where χ m are arbitrary phase coefficients that can assume fixed values or be spatially wound in defects and textures. Crucially, η ∈ [−1, 1] now forms an interpolating parameter between UN and BN magnetic phases: for η = 1, only the ζ 0 component is nonzero, representing the familiar UN phase in Eq. (11) , with the nematic axis aligned with the z axis. Similarly, for η = −1, we retrieve the familiar BN phase in Eq. (12). ...

... Phase vortices penetrating the interface. We first consider a phase vortex in both UN and BN limits, obtained from Eqs. (11) and (12), with k winding ...

Topological interfaces crossed by defects and textures of continuous and discrete point group symmetries in spin-2 Bose-Einstein condensates

Article Full-text available

Jan 2024

Giuseppe Baio · Matthew T. Wheeler · David S. Hall · Janne Ruostekoski · Magnus O. Borgh

We systematically and analytically construct a set of spinor wave functions representing defects and textures that continuously penetrate interfaces between coexisting, topologically distinct magnetic phases in a spin-2 Bose-Einstein…

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... Theoretical investigations have so far been focused mostly on the stability of the interface between distinct sliding fluid components, such as superfluid or normal phases of the same liquid 30,31 or immiscible components in binary Bose-Einstein condensates 32-34 (BECs). Experimental observations remain limited to the interface between the A and B phases of 3 He in a rotating cryostat 35, 36 . All these systems involve phases with distinct physical properties, which complicate the picture due to the presence of additional Rayleigh-Taylor 20 and counter-superflow instabilities 33 . ...

... Theoretical investigations have so far been focused mostly on the stability of the interface between distinct sliding fluid components, such as superfluid or normal phases of the same liquid 30,31 or immiscible components in binary Bose-Einstein condensates [32][33][34] (BECs). Experimental observations remain limited to the interface between the A and B phases of 3 He in a rotating cryostat 35, 36 . All these systems involve phases with distinct physical properties, which complicate the picture due to the presence of additional Rayleigh-Taylor 20 and counter-superflow instabilities 33 . ...

... Our observations establish homogeneous superfluid rings of ultracold fermions as a versatile laboratory for quantum fluid-dynamics experiments. We overcome typical difficulties that complicate experiments with helium quantum fluids 35, 36 , especially in terms of precise initial flow control, suppression of the normal fraction, and single-vortex detection. ...

Universality of the superfluid Kelvin-Helmholtz instability by single-vortex tracking

Preprint

Mar 2023

D. Hernandez-Rajkov · N. Grani · F. Scazza · G. Del Pace · G. Roati

At the interface between two fluid layers in relative motion, infinitesimal fluctuations can be exponentially amplified, inducing vorticity and the breakdown of the laminar flow. This process, known as the Kelvin-Helmholtz instability, is…

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... In superfluids, the Rossby number can be defined in a similar fashion as in classical fluids, while the physical meaning of the Reynolds number is captured by the superfluid Reynolds number Re α (ref. 8), which only depends on intrinsic mutual friction parameters that describe the coupling between the quantized vortices and the normal component 9 . Theoretical 10 , numerical 11 and experimental 12-15 work suggests that, in classical fluids, rotating turbulence, for which Re ≫ 1 and Ro ≪ 1, could be described as an ensemble of interacting inertial waves (IWs) plus a broadband turbulent component, especially for frequencies exceeding twice the angular velocity. ...

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Rotating quantum wave turbulence

Article | Full-text available

Mar 2023 · NAT PHYS

J. T. Mäkinen · S. Autti · P. J. Heikkinen · J. J. Hosio · V. B. Eltsov

Turbulence under strong influence of rotation is described as an ensemble of interacting inertial waves across a wide range of length scales. In macroscopic quantum condensates, the quasiclassical turbulent dynamics at large scales is…

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... The lower panel refers to the uniform region below the vortices, the upper panel to the one above them. Black thin lines are positive-norm modes and red thicker lines are negative-norm ones, that respectively correspond to the plus and minus signs in equation (15). Notice that the horizontal axis is restricted to the first Brillouin zone and the presence in each region of modes of both norms for some values of the Bloch wavenumbers is due to the periodicity of the system along x. ...

... − K, k y = 0), where the frequency is given by the Bogoliubov dispersion relation (15) of sound excitations in a uniform condensate. ...

Interplay of Kelvin-Helmholtz and superradiant instabilities of an array of quantized vortices in a two-dimensional Bose-Einstein condensate

Article | Full-text available

Feb 2023

Luca Giacomelli · Iacopo Carusotto

We investigate the various physical mechanisms that underlie the dynamical instability of a quantized vortex array at the interface between two counter-propagating superflows in a two-dimensional Bose–Einstein condensate. Instabilities of markedly different nature are found to dominate in different flow velocity regimes. For moderate velocities where the two flows are subsonic, the vortex lattice displays a quantized version of the hydrodynamic Kelvin–Helmholtz instability (KHI), with the vortices rolling up and co-rotating. For supersonic flow velocities, the oscillation involved in the KHI can resonantly couple to acoustic excitations propagating away in the bulk fluid on both sides. This makes the KHI rate to be effectively suppressed and other mechanisms to dominate: For finite

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and relatively small systems along the transverse direction, the instability involves a repeated superradiant scattering of sound waves off the vortex lattice; for transversally unbound systems, a radiative instability dominates, leading to the…

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